

It is clear from this table that the proposed modifications of the semi-Lagrangian exponential schemes introduce additional complexity per time step, with two more exponential evaluations and one more spatial interpolation w.r.t. their respective original formulations. However, as assessed in the numerical simulations presented in Section 7, the improved accuracy and stability of SE22 allow us to choose larger time step sizes. Moreover, it has been identified in the numerical simulations that the computation of the exponential functions is responsible for an important fraction of the computing time in our implementation, such that important savings are possible by precomputing them; additional savings could also be obtained with a more efficient implementation (*e.g.*, by using vectorization).

6. LINEAR STABILITY ANALYSIS

In the following paragraphs, we conduct a stability study comparing exponential and semi-Lagrangian exponential schemes, as well as SL-SI-SETTLES. Linear stability analyses of Eulerian exponential schemes have been conducted, *e.g.*, by [3, 8, 9]. The last two cited works show notably that these methods are in general unstable when applied to non-diffusive problems, but depending on configurations of the problem and the exponential scheme, the amplifications are small such that instabilities may remain unnoticed within the simulation time. In this section, we focus on comparing stability regions of both Eulerian and semi-Lagrangian schemes.

The stability study proposed here is inspired by the procedure presented by [8]. Consider a generic nonlinear ODE in the form

$$\dot{u} = \lambda_{\mathbf{L}} u + \mathbf{N}(u) \quad (6.1)$$

Substituting $u = \bar{u} + u'$ in (6.1), where $\bar{u}$ is a steady state (*i.e.*, $\lambda_{\mathbf{L}} \bar{u} + \mathbf{N}(\bar{u}) = 0$) and u' is a perturbation, expanding $\mathbf{N}(\bar{u} + u')$ around $\bar{u}$, truncating at first order and defining $\lambda_{\mathbf{N}} := \mathbf{N}'(\bar{u})$, we obtain the linearized equation

$$\dot{u}' = \lambda_{\mathbf{L}} u' + \lambda_{\mathbf{N}} u' \quad (6.2)$$

which we consider hereafter, with the primes omitted for conciseness. However, we need to take into account that we are comparing both Eulerian (ETDRK) and semi-Lagrangian (SL-SI-SETTLES and SE**) methods, and this study should be able to capture the influence of advective effects and interpolation procedures. In the case of SL schemes, the advection is incorporated into the material derivative, and we can directly use the same linearized ODE as above:

$$\frac{du}{dt} = \lambda_{\mathbf{L}} u + \lambda_{\mathbf{N}} u \quad (6.3)$$

where $u, \lambda_{\mathbf{L}}, \lambda_{\mathbf{N}} \in \mathbb{C}$. This equation arises, *e.g.*, when (3.1) is solved using a spectral method, in which case $u = \hat{u}_k(t)$ is a spectral coefficient of the solution, and $\lambda_{\mathbf{L}}, \lambda_{\mathbf{N}}$ are modes associated respectively to $\mathbf{L}$ and the linearized term $\mathbf{N}$. If $\mathbf{L}$ and $\mathbf{N}$ are diagonalizable in the spectral space with the same set of eigenvectors, then (3.1) leads to a system of uncoupled ODEs of the form (6.3), each one corresponding to a wavenumber k .

On the other hand, in the case of Eulerian schemes, we need to consider the advection explicitly, leading to a linearized one-dimensional scalar advection equation with source term:

$$\frac{\partial u}{\partial t} + v \frac{\partial u}{\partial x} = \lambda_{\mathbf{L}} u + \mathbf{N}(u) \implies \frac{\partial u}{\partial t} + v \frac{\partial u}{\partial x} = \lambda_{\mathbf{L}} u + \lambda_{\mathbf{N}} u \quad (6.4)$$

in which the spatial derivative becomes a scalar multiplication when solved in the spectral space, and v is a scalar velocity field assumed to be constant for simplicity.

The stability region of a given numerical scheme with stability function $A := |u^{n+1}|/|u^n|$ is defined as the region in $\mathbb{C}^2$ in which $|A(\Delta t \lambda_L, \Delta t \lambda_N)| \leq 1$, where $u^n = u(t_n, x) = A^n e^{i\kappa x}$ is the computed approximation to $u(t_n)$, $\kappa = 2\pi k/L$ is the normalized wavenumber and L is the length of the spatial domain. A complete stability study should be performed in a four-dimensional space, since A depends on the real and imaginary parts of λ_L and λ_N ; however, since a hyperbolic problem such as the SWE propagates purely hyperbolic modes, we simplify this study by considering both $\xi_L := \Delta t \lambda_L$ and $\xi_N := \Delta t \lambda_N$ to be purely imaginary (Figure 2), and, in particular, by fixing values of $\xi_L \in i\mathbb{R}$ and plotting the amplification factors as functions of $\Im(\xi_N)$ (Figure 3). We present below the stability functions of each time integration scheme considered in this work.

6.1. Stability of ETD1RK and ETD2RK

When applying (3.5a) and (3.5b) to (6.4), the advection term could be treated either as a linear or nonlinear term, yielding different stability functions. We consider only the latter case, but the overall conclusions on the stability of ETD1RK and ETD2RK compared to the other schemes considered in this work are similar. We have

$$u_{\text{ETD1RK}}^{n+1} = \varphi_0(\xi_L)u^n + \Delta t \varphi_1(\xi_L)\lambda_N u^n - \Delta t \varphi_1(\xi_L)iv\kappa u^n \quad (6.5a)$$

$$= [\varphi_0(\xi_L) + \varphi_1(\xi_L)(\xi_N - i\kappa s)] u^n$$

$$u_{\text{ETD2RK}}^{n+1} = u_{\text{ETD1RK}}^{n+1} + \Delta t \varphi_2(\xi_L) [(\lambda_N - iv\kappa)u_{\text{ETD1RK}}^{n+1} - (\lambda_N - vi\kappa)u^n] \quad (6.5b)$$

$$= u_{\text{ETD1RK}}^{n+1} + \varphi_2(\xi_L)(\xi_N - i\kappa s)(u_{\text{ETD1RK}}^{n+1} - u^n)$$

where $s := v\Delta t = x_j - x_{d_j}$ is the spatial displacement of each particle from t_n to t_{n+1} (see Figure 1). Note that we have considered the spatial derivative to be computed exactly (which can be achieved, *e.g.*, by using a spectral method, as considered in this work), *i.e.*, $(u^n)_x = i\kappa u^n$, since we want to analyze exclusively the stability properties due to the temporal discretization.

Thus, the stability functions of the Eulerian ETD1RK and ETD2RK schemes read

$$A_{\text{ETD1RK}} = \varphi_0(\xi_L) + \varphi_1(\xi_L)(\xi_N - i\kappa s) \quad (6.6a)$$

$$A_{\text{ETD2RK}} = A_{\text{ETD1RK}} + \varphi_2(\xi_L)(\xi_N - i\kappa s)[A_{\text{ETD1RK}} - 1] \quad (6.6b)$$

Note that the presence of the term $-i\kappa s$ makes the stability regions dependent on the spatial information. This implies that the schemes may not be simultaneously stable for a large range of wavenumbers, since the intersection of their stability regions may be small or even empty. Following [17], we compute stability functions $A_{\kappa s} = A_{\kappa s}(\xi_L, \xi_N)$ for $\kappa s \in \mathcal{K} := \{0, \pi/10, 2\pi/10, \dots, 2\pi\}$; then, for each (ξ_L, ξ_N) , the amplification factor is $A(\xi_L, \xi_N) = \max_{\kappa s \in \mathcal{K}} A_{\kappa s}(\xi_N, \xi_L)$.

6.2. Stability of SL-SI-SETTLS

We now develop a linear stability analysis for the semi-implicit semi-Lagrangian SETTLES [17] described in Section 3.2. In the stability analysis of semi-Lagrangian schemes, we also need to take into account the spatial dependence of the solution. Using $u(t_n, x) = A^n e^{i\kappa x}$ in (3.11) applied to (6.3), we obtain

$$A^{n+1}e^{i\kappa x_j} - A^n e^{i\kappa x_{d_j}} = \frac{\xi_L}{2} \left(A^{n+1}e^{i\kappa x_j} + A^n e^{i\kappa x_{d_j}} \right) + \frac{\xi_N}{2} \left([2A^n - A^{n-1}] e^{i\kappa x_{d_j}} + A^n e^{i\kappa x_j} \right) \quad (6.7)$$

which, after division by $A^{n-1}e^{i\kappa x_j}$ and recalling that $s := x_j - x_{d_j}$, leads to the quadratic equation

$$A^2 \left(1 - \frac{\xi_L}{2} \right) - A \left(e^{-i\kappa s} \left(1 + \frac{\xi_L}{2} + \xi_N \right) + \frac{\xi_N}{2} \right) + \frac{\xi_N}{2} e^{-i\kappa s} = 0 \quad (6.8)$$

whose roots $A = A_{\text{SL-SI-SETTLS}}$ define the stability function of SL-SI-SETTLS; its stability region is the intersection between the stability regions defined by each of the two roots of (6.8). Moreover, as in the case of ETD1RK and ETD2RK considering the advection term, these roots depend on the spatial wavenumber through the term $e^{-i\kappa s}$, and the stability region is defined as the intersection of the individual stability regions for each κs .

6.3. Stability of SE11 and SE12

We now study the stability of the semi-Lagrangian exponential schemes proposed by [26] and described in Section 3.3. Using $u(x, t_n) = A^n e^{i\kappa x}$ in (3.15a) and (3.15b) applied to (6.3) and the fact that $\varphi_k(z) = \varphi_0(z)\psi_k(z) \forall k > 0$, we get

$$A_{\text{SE11}} = \varphi_0(\xi_L) e^{-i\kappa s} + \xi_N \varphi_1(\xi_L) e^{-i\kappa s} = e^{-i\kappa s} \varphi_0(\xi_L) [1 + \xi_N \psi_1(\xi_L)] \quad (6.9a)$$

$$\begin{aligned} A_{\text{SE12}} &= A_{\text{SE11}} + \xi_N \varphi_2(\xi_L) [A_{\text{SE11}} - e^{-i\kappa s}] \\ &= e^{-i\kappa s} [(1 + \xi_N \varphi_2(\xi_L)) \varphi_0(\xi_L) (1 + \xi_N \psi_1(\xi_L)) - \xi_N \varphi_2(\xi_L)] \end{aligned} \quad (6.9b)$$

Since $|e^{-i\kappa s}| = 1$, the stability regions of both SE11 and SE12 do not depend on κs . This is a remarkable feature of these schemes, since, contrary to what happens to SL-SI-SETTLS and ETD1RK, there is no reduction of the global stability region when intersecting the regions corresponding to each wavenumber.

We compare in Figure 2 the stability functions of ETD1RK and ETD2RK applied to (6.4), and SL-SI-SETTLS, SE11 and SE12 applied to (6.3), in the plane $(\Im(\xi_N), \Im(\xi_L))$. In Figure 3, we present the amplification factor along two horizontal slices of these planes, with ξ_L chosen as integer multiples of $\tilde{\xi}_L = i\sqrt{\Phi/a^2} \approx 2.5 \times 10^{-4}i$, which is obtained from an approximation to the eigenvalues $\lambda_L = i\sqrt{\Phi n(n+1)/a^2}$ of $L = L_G$ considering relevant physical parameters and an f-plane approximation (see [5] for details). The scaling of $\tilde{\xi}_L$ can be seen as a result of different choices of time step size Δt and/or wavenumbers: for instance, the plot considering $\xi_L = 20000\tilde{\xi}_L$ in Figure 3 corresponds to the stability region of the maximum wavenumber in a spectral resolution $M = 256$ and a reasonable time step size of approximately $\Delta t = 80s$. The plots reveal notably an enhancement of the stability of the semi-Lagrangian exponential schemes when compared to their Eulerian counterparts, which suffer from a small intersection of the stability regions of each κs , indicating that the schemes have a lack of stability common to a large range of combinations between wavenumbers and spatial displacement. The semi-Lagrangian exponential scheme with second-order discretization of the nonlinear term (SE12) also presents a considerably larger stability region compared to SL-SI-SETTLS and SE11, whose stability regions coincide, despite of having different stability functions.